

Urologic Issues for the Nephrologist

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Close interaction between nephrologists and urologists is crucial to the optimal management of several common clinical problems. A proper understanding of urologic strategies helps the nephrologist to ensure that patients with these problems are given clear information and are optimally managed. Areas in which such coordinated work is most important include the management of stone disease, the surgical approach to urinary tract obstruction, the investigation of hematuria, and the management of urinary tract malignant neoplasms.

ADVANCES IN MANAGEMENT OF KIDNEY STONES

Medical management of nephrolithiasis is discussed in [Chapter 60](#). The surgical management of urinary tract stones has been transformed by the introduction of extracorporeal shock wave lithotripsy (ESWL), percutaneous nephrolithotomy (PCNL), and ureteroscopy. These advances mean that open stone surgery is now a final resort when other modalities have been exhausted or are contraindicated. [Table 63.1](#) details the use of different treatment modalities over time since the introduction of newer techniques.¹ As familiarity with the available techniques and technology has developed, standardized treatment strategies have evolved and guidelines have been introduced.^{2,3,4} However, the optimal treatment for certain patients with kidney stones remains controversial, especially lower pole stones.

Improvements in Imaging

Unenhanced computed tomography (CT) scanning of the abdomen and pelvis has replaced intravenous urography (IVU) as the standard imaging modality for stone diagnosis ([Fig. 63.1](#)). CT is readily available, quicker to perform, offers increased sensitivity compared with IVU (99% vs. 70%), and avoids the need for intravenous (IV) contrast. An additional advantage is that CT can demonstrate radiolucent stones (mainly uric acid and xanthine stones) and detect concomitant lesions and/or alternative diagnoses. CT requires an increased radiation dose, but this is less of an issue with modern equipment and newer radiation protocols. Comparative doses are 2.5 mSv for IVU, 5 mSv for standard non-contrast-enhanced CT, and 2 mSv for low-dose non-contrast-enhanced CT.

Conservative (Nonsurgical) Management

Spontaneous stone passage can be expected in up to 80% of patients with stones smaller than 4 mm. Conversely, for stones with a diameter of more than 7 mm, the chance of spontaneous stone passage is very low. Recent data from the MIMIC study, a multicenter retrospective cohort study of 4170 patients in 71 institutions across four countries, suggests a chance of just 29%.⁵ The location is also important; up to 70% of distal ureteral stones pass spontaneously, in contrast to only 45% of midureteral and 25% of proximal ureteral stones. Intervention is recommended when there is persistent pain (>72 hours) despite

adequate analgesia, persistent obstruction with risk for impaired kidney function (e.g., with preexisting kidney impairment or in a single kidney), bilateral obstruction, or associated urinary tract sepsis.

In the absence of an acute indication for surgical management, medical expulsive therapy (MET), such as tamsulosin (an alpha blocker) 400 µg once daily or nifedipine (calcium channel blocker) 30 mg once daily, is sometimes used to aid stone passage, particularly for distal ureteral stones, although the data on the efficacy of this are mixed. A recent U.K. study (the largest randomized controlled trial [RCT] of MET) demonstrated no significant benefit of this therapy.⁶ This was supported by the MIMIC study.⁵ In contrast, data from several meta-analyses of RCTs suggest that MET is beneficial for management of distal ureteral stones less than 10 mm.^{4,7,8} Therefore, MET for small distal ureteral stones is still recommended in most guidelines.^{2,3,4} Opinion among urologists remains divided.

Another conservative approach is chemolysis because several stone types are, in principle, amenable to dissolution by oral medications or by direct instillation of chemical solutions. However, chemolysis is only effective for uric acid stones and some drug-induced calculi, which can be readily dissolved by alkalization of the urine, usually with oral potassium citrate, or with sodium bicarbonate solution instilled directly into the urinary tract via a percutaneous nephrostomy (PCN) tube.

Acute Surgical Intervention

The goals of acute surgical intervention are to relieve obstruction and, if feasible, to remove the calculus. If the patient is well enough for general anesthesia, ureteroscopic stone destruction using laser lithotripsy can be attempted. Alternatively, a double-J stent (a ureteral stent with two coiled ends) can be inserted to relieve obstruction until definitive treatment is performed ([Fig. 63.2](#)). Acute ESWL is increasingly being used for ureteral stones in stable patients with adequate pain relief and without superimposed infection. However, in the setting of uncontrolled urinary tract infection (UTI) resulting from an obstructing stone, decompression of the urinary tract and free drainage of urine are the priorities. Evidence suggests little difference in the outcomes whether an antegrade (i.e., PCN) or retrograde (i.e., double-J stent) is used.⁹ However, PCN is generally the preferred option (when not contraindicated) because it can be performed with local anesthesia and is less likely than endoscopic surgery to cause bacteremia ([Fig. 63.3](#)).

Elective Surgical Intervention

Extracorporeal Shock Wave Lithotripsy

During ESWL, acoustic shock wave energy is delivered to a stone under fluoroscopic and/or ultrasound guidance. Treatment sessions typically last about 30 minutes, during which 1500 to 2500 shock waves are delivered. Treatment is given to outpatients under analgesia or IV sedation and can be repeated at intervals of 10 to 14 days. Stones up

TABLE 63.1 Changing Use of Techniques for Stone Removal

	1984	1990	1999
Location (%)			
Calyceal stones	35	43	46
Pelvic stones	42	20	13
Staghorn stones	8	3	1
Ureteral stones	15	34	40
Treatment Modality (%)			
ESWL	60	79	78
PCNL	20	5	2
Ureteroscopy	11	15	20
Open surgery	9	1	0.1

ESWL, Extracorporeal shock wave lithotripsy; PCNL, percutaneous nephrolithotomy.

Data from Rassweiler JJ, Renner C, Eisenberger F. The management of complex renal stones. *BJU Int.* 2000;86:919–928.



Fig. 63.1 Computed Tomography Scan Demonstrating a Ureteral Calculus. Non-contrast-enhanced scan showing a calculus (arrow) at the right vesicoureteral junction.



Fig. 63.2 Ureteral Stenting. Plain radiograph showing a double-J stent in the left ureter. Note that the curled ends of the stent remain in the pelvis despite ureteral peristalsis.



Fig. 63.3 Nephrostogram in Ureteral Obstruction Caused by a Stone. Contrast material is injected through a percutaneous nephrostomy tube placed in the lower pole calyx (arrow). The contrast material outlines a single large calculus (arrowheads) producing complete obstruction at the pelviureteral junction.

to 20 mm in size can be treated effectively, and stone-free rates of 60% to 98% have been reported. However, ESWL is operator dependent, and outcome is influenced by the size, composition, and location of the stone and the type of lithotripter used. Cystine and calcium oxalate monohydrate stones are especially resistant. Targeting of the stone may be impossible in the presence of obesity and skeletal deformities (increased skin-to-stone distance), and ESWL is contraindicated in patients with aortic or renal artery aneurysm, uncontrolled UTI, coagulation disorders/receiving anticoagulation treatment, pregnant persons, and obstruction distal to the stone being treated.

A double-J ureteral stent is sometimes placed endoscopically before ESWL treatment to prevent stone fragments from obstructing the distal ureter (*Steinstrasse*, literally “stone street”; Fig. 63.4). Other acute complications of ESWL include hemorrhage or hematoma, infection, and injury to adjacent organs. The risk for hypertension or reduced glomerular filtration rate (GFR) as sequelae of ESWL is controversial.

Percutaneous Nephrolithotomy

During PCNL, a tract through the kidney parenchyma is created between the skin and the collecting system of the kidney. A sheath within this tract is used as a working channel to remove stones under direct vision using a nephroscope. Preoperatively, CT imaging is used to localize calculi and neighboring organs (e.g., spleen, liver, large bowel, pleura, or lungs) and to plan access. The most frequently used access site is the dorsal calyx of the lower pole, and stone fragmentation is undertaken by ultrasound, pneumatic, or laser devices. With technologic advances, performing this procedure through smaller access sheaths (<18 Fr compared with standard 30 Fr sheaths) with fewer complications and equivalent stone clearance rates is becoming feasible.¹⁰ The PCNL technique is modified for special circumstances, usually by altering the site of puncture (e.g., directly into a calyceal diverticulum) or, if there are ureteral stones, by using a higher placed puncture to permit antegrade ureteroscopy. The percutaneous puncture may be facilitated by the preliminary placement of a retrograde ureteral catheter to dilate and opacify the collecting system, which is then punctured under fluoroscopy with or without ultrasound

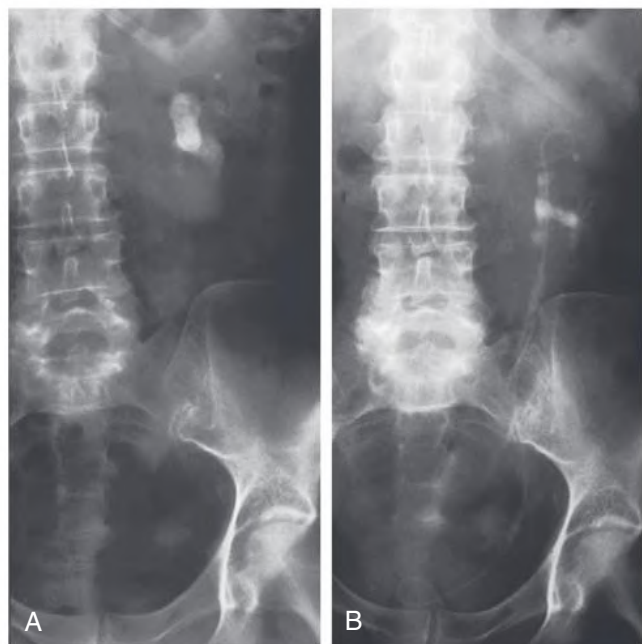


Fig. 63.4 Extracorporeal Shock Wave Lithotripsy (ESWL) Complicated by Stone Fragments Obstructing the Distal Ureter. (A) Preoperative plain radiograph showing stones in the left kidney pelvis. (B) After ESWL, note the disappearance of the pelvic stone, the string of stone fragments throughout the length of the ureter, and the double-J ureteral stent placed to facilitate their passage.

guidance. After completion of PCNL, there are a number of options to aid urinary drainage. These include ureteral stents (placed antegradely or retrogradely), “pig-tail” nephrostomy tubes, or a self-retaining balloon nephrostomy tube to tamponade the tract and provide further access if needed. Hemorrhage can complicate PCNL (from intrarenal or, rarely, intercostal/subcostal arteries) and usually can be treated conservatively or by selective angiographic embolization. Other complications include sepsis; fluid overload (similar to transurethral resection syndrome); injury to spleen, pleura, or colon; and extravasation. PCNL usually results in minimal parenchymal injury, averaging only 0.15% of the total kidney cortex.¹¹

Indications for PCNL are shown in Table 63.2. These continue to evolve and are being challenged by developments in ureteroscopic techniques, which are allowing more upper ureteral and renal pelvic stones to be dealt with by a retrograde approach.

ESWL is the first-line treatment for more than 75% of stone patients that require an intervention. Table 63.2 shows circumstances in which ESWL is less effective and PCNL becomes the preferred approach or a combination of the two modalities is used. For lower pole stones in particular, ESWL may not provide optimal clearance because of problems with the drainage of residual fragments. An RCT has shown that for lower pole stones greater than 10 mm, PCNL has much better clearance rates than ESWL (92% vs. 23%).¹²

Open Stone Surgery

Open surgery still has a place in the treatment of stone disease. Approximately 2% of stone patients are now treated with open surgery, mainly when anatomic factors preclude the use of minimally invasive methods or when these techniques have failed. Other indications include complex stone burden, the presence of intrarenal anatomic abnormalities (e.g., pelviureteral junction [PUJ] obstruction), and stones within a nonfunctioning kidney where nephrectomy may

TABLE 63.2 Indications for Percutaneous Nephrolithotomy

Composition ^a	Struvite stones	Complete removal necessary to eliminate infection and minimize stone recurrence
	Calcium oxalate monohydrate stones	Difficult to pulverize by ESWL
	Cystine stones	Difficult to pulverize by ESWL
Stone position	Lower pole stones	Fragments less easily evacuated from dependent lower pole calyces, especially if collecting system dilated
Anatomic abnormalities	PUJ obstruction Calyceal diverticula	Prevent passage of fragments after ESWL
Patient characteristics	Morbid obesity Ureteral obstruction	Stone cannot be placed in focal point of ESWL machine

ESWL is the first choice for stone intervention, except in those circumstances that may favor PCNL.

^aStone composition can be defined with certainty only by direct stone analysis, but advances in imaging may ultimately provide a means to accurately assess stone composition in situ before treatment, thus allowing the urologist to select the treatment most likely to be successful.

ESWL, Extracorporeal shock wave lithotripsy; PCNL, percutaneous nephrolithotomy; PUJ, pelviureteral junction.

be more appropriate. During surgery the kidney pelvis and the parenchyma can be opened along avascular planes, and clamping of the kidney vessels and hypothermia of the kidney may be needed. In selected patients a laparoscopic or robot-assisted approach can be used for the treatment of stone disease.

Ureteroscopy

Continued advances in the design of endoscopes for ureteronephroscopy have rendered the entire urinary tract accessible to endoscopic examination and manipulation. Ureteroscopes may be semirigid or flexible, the latter allowing access to the renal pelvis and calyces. Stone fragmentation is achieved ideally by laser but also by ultrasound or pneumatic devices (lithoclast). Laser use is equally effective for all types of stones and has the additional advantages of a flexible fiber (allowing intrarenal stone fragmentation), low tissue penetration, and minimal stone displacement during use. Success rates for laser fragmentation of ureteral stones are approximately 80%. Fig. 63.5 highlights the increasing use of ureteroscopy in stone disease and also the increasing proportion of procedures in which a laser rather than ultrasound or a lithoclast is used to achieve stone fragmentation.

For lower pole kidney stones, although flexible ureteroscopes can access the lower pole to facilitate treatment, the stone-free rate in comparative studies is similar to that of ESWL.^{13,14} Furthermore, patients often prefer ESWL as the initial therapy compared with flexible ureterorenoscopy and lasertripsy. In the event of failed stone clearance after ureteroscopy, PCNL is commonly used next and is likely to remain an essential treatment for lower pole stones. Complications of ureteroscopy, particularly with use of graspers and baskets, include ureteral avulsion, perforation, extravasation, mucosal damage, hematuria, infection, and stricture. Advances in laser technology now enable stones to be reduced to dust-like particles, reducing the need for graspers and baskets and hence reducing complications.

A summary of the elective surgical management of nonobstructing kidney stones is shown in Table 63.3.

Changing Use of Endoscopy to Achieve Stone Fragmentation

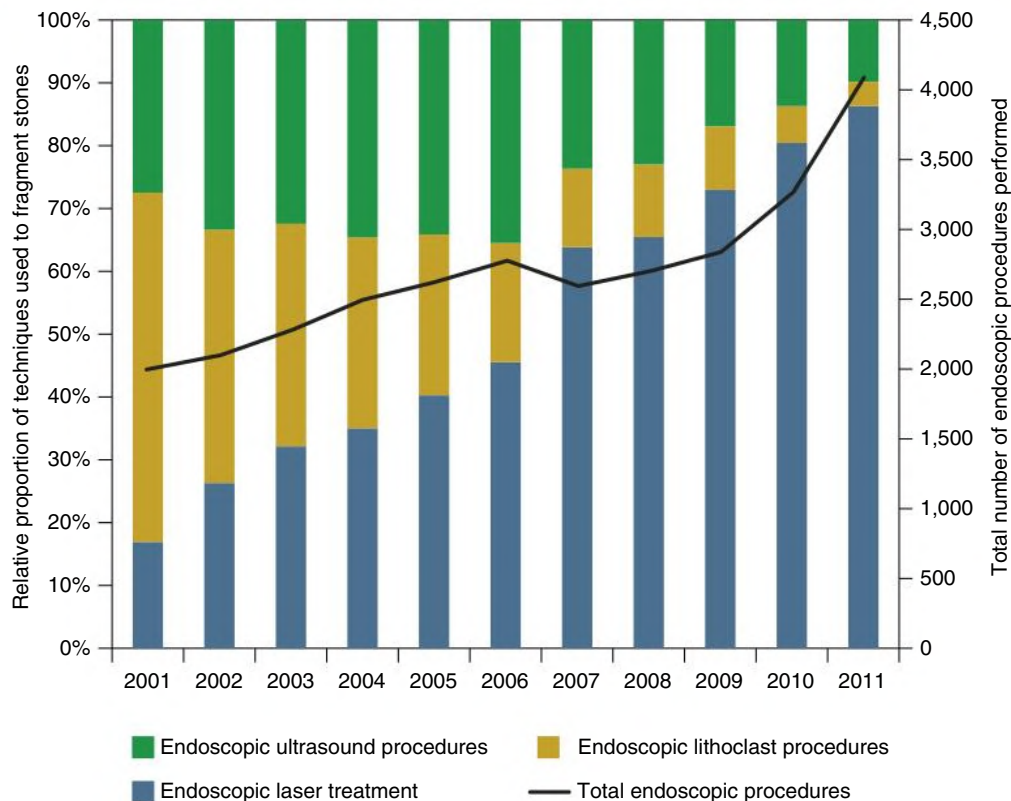


Fig. 63.5 Changing Use of Endoscopy to Achieve Stone Fragmentation. Chart showing the increasing use of endoscopy in stone surgery in addition to the increasing proportion of cases using lasers to achieve stone fragmentation. (Data from UK Hospital Episode Statistics dataset, <http://www.hscic.gov.uk/hes>.)

TABLE 63.3 Surgical Management of Intrarenal Stones

Stone Characteristic	First-Line Treatment Modality
Nonlower pole stone <20 mm	ESWL or URS
Nonlower pole stone >20 mm	PCNL
Lower pole stone <10 mm	ESWL or URS
Lower pole stone >10 mm	PCNL or URS

ESWL, Extracorporeal shockwave lithotripsy; PCNL, percutaneous nephrolithotomy; URS, ureteroscopy.

Data from Assimos D, Krambeck A, Miller N, et al. American Urological Association/Endourological Society Guideline: Surgical management of Stones, 2016. <https://www.auanet.org/guidelines/kidney-stones-surgical-management-guideline>.

Management of Staghorn Calculus

A staghorn calculus usually should be managed by intervention because reports of conservative therapy show a high rate of eventual nephrectomy (up to 50%) and an increase in associated morbidity (mainly kidney failure) and mortality (up to 28%). Patient age and GFR heavily influence treatment decision making. The primary treatment option for staghorn calculi in a patient with preserved kidney function is PCNL. For a staghorn calculus in a nonfunctioning kidney, partial or total nephrectomy may be indicated. ESWL or ureteroscopic stone removal alone are rarely effective for large staghorn calculi but

can be used as an adjunct to PCNL to achieve total stone clearance. The advantage of a multimodal approach is the reduced need for additional access to the kidney and secondary PCNL. Consequently, endoscopic combined intrarenal surgery (ECIRS; i.e., retrograde flexible ureterorenoscopy and PCNL) is becoming more commonplace. Very rarely, open surgical removal of the stone may be indicated.

Stones in Transplanted Kidneys

The management of stone disease in a transplanted kidney is challenging because of the solitary kidney, the anatomic location within the pelvis, and the difficulty with retrograde access to the ureter and kidney. Early active intervention is indicated; prophylactic stenting, ureteroscopy, and PCNL are preferred to ESWL because stone targeting may not be possible. Open surgery may be needed in selected cases.

URINARY TRACT OBSTRUCTION

General Aspect

The causes of upper tract obstruction are listed in [Chapter 61](#) (see [Box 61.2](#)), as is a summary of the management of obstruction (see [Fig. 61.7](#)). Upper tract obstruction caused by malignancy can be a result of direct tumor invasion, external compression by metastatic lymph node involvement or, rarely, true metastasis to the ureter. About 70% of tumors causing ureteral obstruction are genitourinary (cervical, bladder, prostate) in origin; breast and gastrointestinal carcinomas and lymphoma constitute the majority of the remainder.¹⁵ The

presentation of a patient with obstruction may vary significantly, and hydronephrosis may develop progressively and insidiously and remain unrecognized until the patient develops anuria and uremia. Upper tract obstruction from malignancy rarely manifests with classic acute ureteral colic, which is typically seen with a benign cause such as a stone. Therefore, these patients often present first to a nephrologist because of an asymptomatic decline in GFR.

Bladder outflow obstruction may be acute and dramatic or chronic with few or no symptoms. To aid in understanding of the immediate needs and longer-term prognosis of these patients, the acute and the definitive management of the most common urologic diseases associated with obstruction and reduced GFR are outlined.

Acute Management

Relief of obstruction is crucial to reverse kidney impairment and preserve remaining kidney function. In cases of bladder outflow obstruction, a urethral or suprapubic catheter is indicated, whereas in upper tract obstruction, a double-J stent is preferable, when possible. The most straightforward approach is endoscopic retrograde placement under fluoroscopy, with PCN reserved for patients in whom the procedure fails. Bilateral stents should be placed if technically possible. However, tumor infiltration can distort trigonal anatomy, making identification of ureteral orifices for double-J stent insertion impossible at the time of cystoscopy. Furthermore, it has been suggested that stents fail to relieve obstruction in 40% to 50% of cases of external ureteral compression. Thus, these patients need to be closely monitored to ensure resolution of the obstruction. Alternatives include placing two stents within each obstructed ureter¹⁶ and a metallic, self-expanding stent, used alone or in conjunction with double-J stents, which has had good initial results in maintaining ureteral patency and avoiding PCN, in malignant ureteral obstruction.¹⁷

A stable patient with obstruction but without signs of sepsis is a candidate for attempted retrograde stent placement with use of general anesthesia. However, in a patient with sepsis, endoscopic manipulation can lead to bacteremia and septic shock. Furthermore, such patients may not be fit for general anesthesia, in which case the preferred initial approach is PCN, which can then be followed, after an interval, with antegrade ureteral stenting. The success rate of this combined approach is high (>90%).¹⁸ In patients with bilateral ureteral obstruction, it is not always necessary to insert bilateral PCN tubes. Significant palliation and return to nearly normal GFR can be accomplished by drainage of the kidney with the better-preserved parenchyma as determined by CT scan or ultrasound. Once they have been placed, PCN tubes or double-J stents need to be replaced every 3 to 6 months. If they are left for a longer period, they become increasingly brittle and encrusted and are liable to block or fracture under manipulation. Complications of ureteral stents include migration, obstruction with proteinaceous material, infection, fragmentation, and, rarely, erosion through the urinary tract.¹⁹ As many as 70% of patients with stents report lower urinary tract symptoms, mainly urgency, frequency, and nocturia, as well as pain along the urinary tract.

Morbidity after stenting or PCN is similar.²⁰ The main problem with indwelling stents is the increased risk for recurrent obstruction (11% for stents vs. 1% for PCN). PCN may have an increased infection rate, and there may be psychological issues relating to the need for an external drainage bag. They are also more liable to displace. An alternative to attempt to offset this is the nephroureteral stent (otherwise referred to as a nephroureterostomy).²¹ Placed antegradely, a single stent with a coil in the bladder and further coil in the kidney pelvis then continues and exits percutaneously to drain into an external nephrostomy bag.



Fig. 63.6 Extraanatomic Stenting for Malignant Ureteral Obstruction. Plain radiograph showing placement of an extraanatomic stent for malignant obstruction of the right ureter. The upper end of the double-J stent has been placed in the right kidney pelvis (arrow). The stent then runs through a subcutaneous tunnel before the lower end enters the bladder (arrowhead).

Extra-anatomic stents are an alternative for patients in whom conventional stent insertion has failed or for whom permanent nephrostomy drainage is unacceptable. An extra-anatomic stent is placed by an initial percutaneous puncture and insertion of the upper end of a long (50-cm) double-J stent into the kidney. A subcutaneous tunnel is then created to bring the stent to the level of the iliac crest. Another tunnel is fashioned to bring the lower end of the stent out suprapubically, followed, finally, by suprapubic puncture of a full bladder and insertion of the lower end (Fig. 63.6).²² Extra-anatomic stents are usually changed at 6-month intervals, and preliminary experience confirms their value in maintaining ureteral patency and avoiding PCN. Because of the effectiveness of minimally invasive methods, open surgery today is rarely indicated in the acute setting.

SPECIFIC TYPES OF OBSTRUCTION

Pelviureteral Junction Obstruction

Surgical management of PUJ obstruction is aimed at obliterating the redundant (aperistaltic) portion of the ureter, allowing normalization of drainage from the affected kidney. The need for treatment is heralded by the onset of pain, infection, calculus formation, decline in GFR, or proven impaired excretion on mercaptoacetyl triglycine (MAG3) renography. The gold standard treatment is now laparoscopic or robotic pyeloplasty, although traditional open techniques are still used. Endoscopic (endopyelotomy) treatments are also performed in some centers. However, in treatment-naïve patients, endopyelotomy has inferior outcomes compared with pyeloplasty.^{23,24}

Retroperitoneal Fibrosis

Retroperitoneal fibrosis can be treated medically or surgically. A causative factor should be excluded (see Chapter 61). There are reports of favorable outcomes after immunosuppression with high doses of corticosteroids or azathioprine/mycophenolate mofetil/rituximab. Alternatively, surgery, consisting of ureterolysis (freeing the ureters

from the fibrotic plaques) and omentoplasty (transposition of the ureters into the peritoneal cavity in omental wraps), can have good long-term results.

Malignant Obstruction

For upper tract transitional cell carcinoma, acute obstruction is best treated by internal stenting. PCN is avoided because of the risk of tumor seeding. Upper tract transitional cell carcinoma is an aggressive tumor and necessitates prompt extirpative surgery.

Bladder cancer can lead to hydronephrosis by invading the ureteral orifices and intramural ureters. In the absence of metastatic disease, radical cystectomy is indicated, although temporary ureteral decompression may normalize GFR if neoadjuvant chemotherapy is planned. In selected cases, bladder preservation strategies combining systemic chemotherapy and radiotherapy can offer long-term control of the disease.

Prostate cancer causes obstruction by occluding the urethra or invading the ureteral orifices. Hormonal treatment can shrink prostatic tissue and malignant deposits and offer long-term remission of symptoms. Limited resection of the prostate may sometimes be indicated.

The decision to offer ureteral decompression for upper tract obstruction caused by cancer is not straightforward and requires input not only from the urologist but also from colleagues in radiation and medical oncology and the palliative care team. There must also be careful discussion of the options with the patient and family.

Ureteral decompression is justified when radiotherapy and systemic chemotherapy remain therapeutic options after improvement in GFR but also may be justified for palliation of pain or ongoing renal tract sepsis.

A review of patients undergoing PCN for obstructive uropathy secondary to pelvic malignancy identified a subset with very poor survival in whom ureteral decompression is usually not justified (Table 63.4).²⁵ Patients with gastric or pancreatic cancer survive a median of only 1.4 months after ureteral decompression.²⁶ In another report, the average survival of patients with advanced malignant neoplasms undergoing endourologic diversion was only 5 months, half of which time was spent in the hospital.²⁷ A more recent review suggests a median survival of 6.4 months for all-cause malignant ureteral obstruction and a complication rate as high as 41%.²⁸

Benign Ureteral Strictures

These can be secondary to stone disease, iatrogenic, or caused by various benign diseases. The treatment of choice is endoscopic balloon dilation or ureterotomy. Open surgical repair or major reconstructive surgery may be needed in cases of recurrent strictures.

Hydronephrosis of Pregnancy

Hydronephrosis occurs in the majority of pregnant persons, typically seen from 6 to 10 weeks of gestation onward. It is caused by both mechanical obstruction from the gravid uterus and dilated ovarian venous plexus and a reduction in peristalsis caused by increasing levels of progesterone leading to smooth muscle relaxation. Resolution typically occurs within a few weeks of delivery but can take up to 3 months. Females are usually asymptomatic with little or no impact on GFR, but they can provide a diagnostic dilemma if presenting with loin pain. Differentiating between a physiologic hydronephrosis and one caused by ureteral obstruction secondary to a stone can be a challenge because the use of ionizing radiation is to be avoided.

Bladder Outflow Obstruction

Bladder outflow obstruction in males is most frequently caused by either benign or malignant prostatic enlargement. Common presentations of

TABLE 63.4 Percutaneous Nephrostomy for Malignant Obstructive Uropathy*

	Median Survival (wk)	5-Year Survival Rate (%)
Group I: primary untreated malignancy	27	10
Group II: recurrent malignancy with further treatment	20	20
Group III: recurrent malignancy with no further treatment	6.5	None survived >1 year
Group IV: benign disease as a result of previous treatment	Not stated	64
Overall	26	22

*Outcome in 77 patients undergoing percutaneous nephrostomy for obstructive uropathy secondary to pelvic malignant disease. Data from Lau MW, Temperley DE, Mehta S, et al. Urinary tract obstruction and nephrostomy drainage in pelvic malignant disease. *Br J Urol.* 1995;76:565–569.

both conditions include the onset of lower urinary tract symptoms and acute (painful) or chronic (painless) retention of urine.

Chronic retention can be considered as the maintenance of voiding with incomplete bladder emptying. It is further classified into low- and high-pressure chronic retention. Low-pressure chronic retention occurs in the absence of upper tract compromise, whereas high-pressure chronic retention is associated with hydronephrosis and kidney injury.

In acute retention, low-pressure chronic retention, and lower urinary tract symptoms caused by bladder outflow obstruction, treatment with 5 α -reductase inhibitors (finasteride, dutasteride) or α -adrenergic receptor blockers (tamsulosin, alfuzosin) is usually indicated before surgery is offered.²⁹ Given the risk for worsening kidney failure if medical therapy fails, it is not a safe management option in high-pressure chronic retention. Surgical management is reserved for patients who either derive no benefit from or are not willing to pursue medical therapy, those with acute and low-pressure chronic retention, and those with high-pressure chronic retention. Before surgery, evaluation of likelihood of success is essential. Patients with very high postvoid residual volumes of urine (>1 L) are less likely to benefit from surgery because of the chronicity of their symptoms and consequent detrusor muscle weakness. These patients are often treated best with intermittent clean self-catheterization or a permanent urethral or suprapubic catheter.

The number of surgical management options available for bladder outflow obstruction is continually increasing. Although transurethral resection of the prostate (TURP) has been the gold standard procedure for decades, the development of laser technology is challenging this. Holmium laser enucleation of the prostate (HoLEP) and vaporization of prostatic tissue (Green light laser) are being increasingly performed in specialist centers with superior results with respect to reduced patient stay, bleeding, and duration of postoperative catheterization. Unlike standard TURP, laser prostatectomy also can be routinely performed safely in patients with large gland size (>100 mL) and on anticoagulation therapy. A recently published 7-year follow-up study comparing TURP with HoLEP reported no significant difference in quality of life, maximum urinary flow, incontinence, or erectile dysfunction, with a reduced need for reoperation in the HoLEP arm of the study.³⁰ More novel techniques aimed at being less invasive and reducing the impact on sexual function and continence are becoming more commonplace. Examples include, but are not limited to, UroLift

(urethraly placed implants that tack the occluding prostate out of the lumen),³¹ Aquablation (robot-assisted waterjet),³² and Rezum (water vapor treatment).³³

Neurologic Diseases of the Lower Urinary Tract

Diseases of the central or peripheral nervous system can manifest with bladder underactivity, bladder overactivity, and/or detrusor-sphincter dyssynergia and lead to bilateral hydronephrosis. Diabetes mellitus also can produce a flaccid denervated bladder through destruction of the peripheral nerves and can cause chronic retention and kidney failure. Of diabetics who develop peripheral neuropathy, 75% to 100% will develop some neurogenic lower urinary tract dysfunction.

The goal of treatment in patients with neurogenic bladder dysfunction is to maintain low intravesical pressures to protect the kidneys. This can be achieved in a number of ways including clean intermittent self-catheterization, long-term suprapubic catheterization, anticholinergics, cystoscopic injection of intradetrusor onabotulinum toxin-A, or surgery, including augmentation cystoplasty.

INVESTIGATION OF HEMATURIA

Macrohematuria is perhaps the most important symptom in urologic practice, and apart from being alarming to the patient, it can be the first presenting sign of an underlying malignant urologic condition (most often a transitional cell tumor of the bladder). Studies show that 15% to 22% of patients with visible hematuria have an underlying genitourinary tract malignant neoplasm.

Patients with visible hematuria must be distinguished from those who have been found to have microhematuria (dipstick), in whom the risk for malignant change is significantly lower (2%–11%).

The outcome of full evaluation of a large group of patients attending a hematuria clinic is shown in Table 63.5.³⁴ In addition to a small but important group of patients in whom malignant disease was identified, there was a significant pickup rate of parenchymal kidney disease (~10%) presenting with both visible and nonvisible hematuria. It is also important to note the sizable proportion of patients in whom a definitive diagnosis could not be reached.

Evaluation of Visible Hematuria

All adults with a single episode of visible hematuria require full urologic evaluation, including kidney imaging and cystoscopy. The only exception to this rule occurs when an adult younger than 40 years gives a history characteristic of glomerular hematuria, such as is typically seen in immunoglobulin A (IgA) nephropathy, in which dark brown hematuria lasting 24 to 48 hours coincides with intercurrent mucosal infection, usually of the upper respiratory tract. This hematuria may be painless, or there may be bilateral loin ache. These young adults should be referred first for nephrologic assessment.

Evaluation of Microhematuria

Perhaps the greatest degree of overlap in practice between urologists and nephrologists occurs in the initial assessment of patients with microhematuria. Concurrent UTI will cause microhematuria and should be treated before further evaluation. Furthermore, features such as menstruation, vigorous exercise, viral illness, and trauma can themselves account for microhematuria. In the presence of a transient or treatable underlying cause other than UTI, repeat urinalysis should be performed 2 days later. Further urinalysis should be deferred until 6 weeks after treatment of confirmed UTI. If microhematuria resolves, no further investigation is required.

The 2012 American Urological Association (AUA) guidelines on the assessment and management of asymptomatic microscopic

TABLE 63.5 Outcome of Evaluation in a Hematuria Clinic: Cases Investigated

Diagnoses Found	All (%)	Microscopic Hematuria (%)	Macroscopic Hematuria (%)
No diagnosis	63	69	53
Kidney cancer	0.6	0.3	0.9
Upper tract transitional cell carcinoma	0.1	0.1	0.1
Bladder cancer	12	5	19
Prostate cancer	0.4	0.2	0.6
Stone disease	4	4	3
Urinary tract infection	13	13	13
Kidney parenchymal disease	10	9	10
Likelihood of Finding Malignancy			
Male, age >40 yr	8		24
Male, age <40 yr	2		7
Female, age >40 yr	5		19
Female, age <40 yr	0		0

Data from Khadra MH, Pickard RS, Charlton M, et al. A prospective analysis of 1,930 patients with hematuria to evaluate current diagnostic practice. *J Urol*. 2000;163:524–527.

(nonvisible) hematuria (AMH) defined AMH by the presence of three or more red blood cells (RBCs) per high-power field (HPF) on microscopic examination of one properly collected, noncontaminated urine sample. In addition to the benign causes listed, pyuria and bacteriuria also should be excluded with either dipstick or microscopy.

The most notable departure from previous guidance is the recommendation of urologic investigation after one episode of AMH, rather than waiting for two of three positive samples as previously suggested. No studies directly compare the diagnostic yield of investigating the first episode of AMH rather than reserving investigation for patients with more than one episode. However, indirect comparison reveals a diagnostic yield of urinary tract malignancy of 1.8% and 3.6% in patients with one or more than one episode of AMH, respectively.³⁵ Dipstick-positive hematuria may still herald significant disease in the absence of RBCs on microscopy because RBCs may lyse in alkaline or hypotonic urine before reaching the laboratory for analysis.³⁶

Complete evaluation of microhematuria includes a history and physical examination, laboratory analysis, and radiologic imaging of the upper urinary tract, followed by cystoscopy (Fig. 63.7). The 2020 update to the AUA guidelines introduces risk stratification to the workup of patients with microhematuria. This provides guidance as to what imaging modality of the upper tract should be performed.³⁷ In females, urethral and vaginal examinations should be performed to exclude local causes of microhematuria. In uncircumcised males, the foreskin should be retracted to expose the glans penis, if possible. If phimosis is present, a catheter specimen of urine may be required. An assessment of GFR should be made (serum creatinine, estimated GFR [eGFR]) because intrinsic kidney disease may have implications for consequent imaging and treatment. The remaining laboratory investigations are guided by specific findings of the history, physical examination, and urinalysis. Multiphase CT urography, which allows evaluation of kidney parenchyma and also the collecting system, has been shown to have the highest sensitivity and specificity for imaging the upper tracts. If CT urography is unavailable, then ultrasound is an alternative imaging strategy. The 2020 guidelines recommend cystoscopy for all patients with AMH who are at least AUA intermediate risk

Evaluation of Asymptomatic Microscopic Hematuria

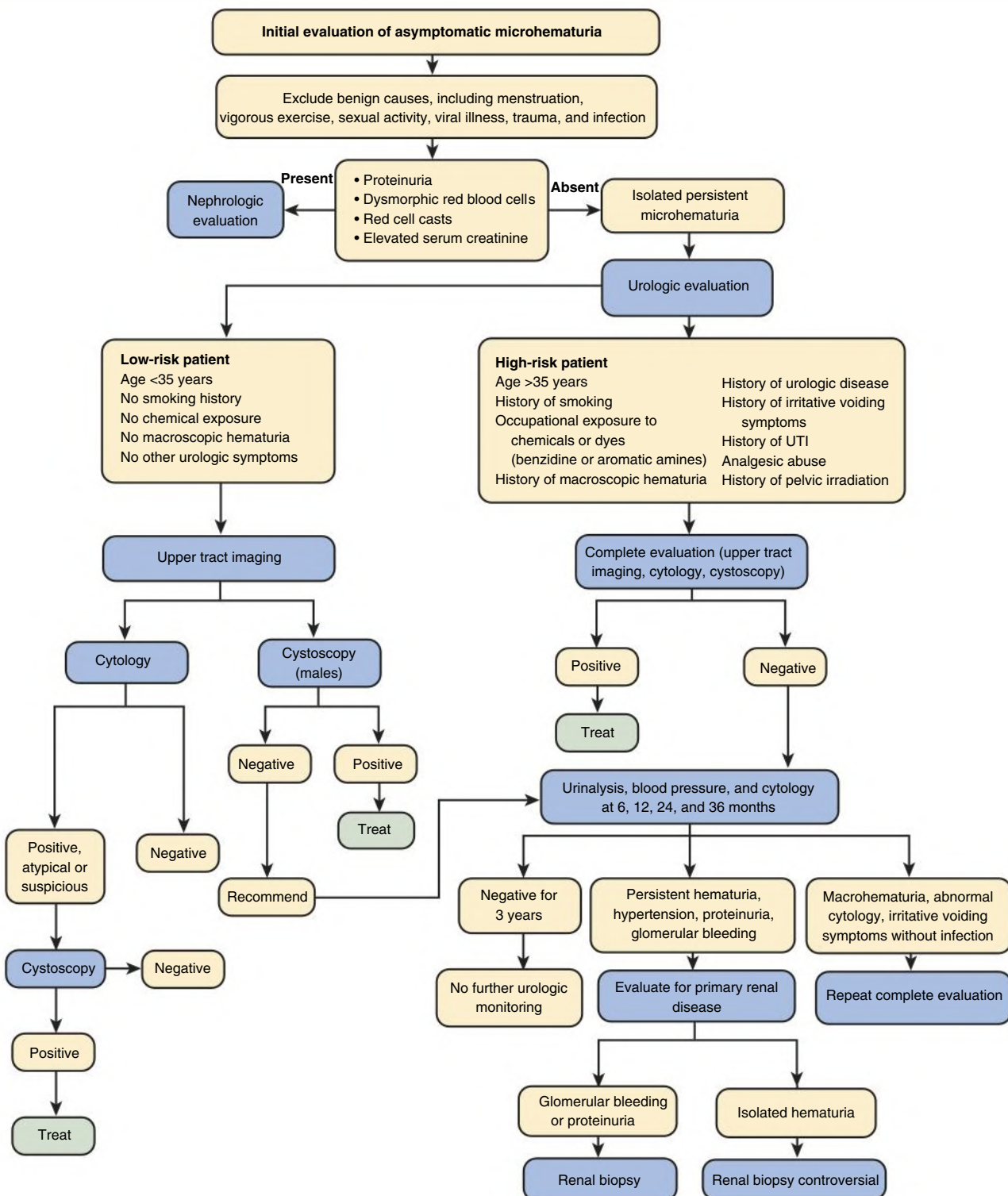


Fig. 63.7 Evaluation of asymptomatic microscopic hematuria. UTI, Urinary tract infection.

for a urologic malignancy (i.e., females >50 years old, males >40 years old, those with a more than 10 pack per year smoking history, >11 RBCs per HPF, recurrent AMH, or with specific risk factors for urologic malignancy or clinical suspicion of malignancy) (Box 63.1). The

use of adjuncts to AMH workup such as urine cytologic examination and other urine biomarkers is no longer advocated.³⁷

Significant proteinuria (>0.3 g/24 h), RBC casts, predominance of dysmorphic RBCs in the urine, or reduced GFR should prompt

BOX 63.1 Common Risk Factors for Urinary Tract Malignancy in Patients With Microhematuria

- Male sex
- Age (>35 years)
- Past or current smoking
- Occupational or other exposure to chemicals or dyes (benzenes or aromatic amines)
- Analgesic abuse
- History of visible hematuria
- History of urologic disorder or disease
- History of irritative voiding symptoms
- History of pelvic irradiation
- History of chronic urinary tract infection
- History of exposure to known carcinogenic agents or chemotherapy such as alkylating agents
- History of chronic indwelling foreign body

referral to a nephrologist and evaluation for parenchymal kidney disease. When present, RBC casts are virtually pathognomonic of glomerular bleeding, but they are often absent in low-grade glomerular disease. Accurate determination of RBC morphology requires inverted phase-contrast microscopy. In general, glomerular bleeding is associated with more than 80% dysmorphic RBCs, and lower urinary tract bleeding is associated with more than 80% normal RBCs.³⁸ This assessment is operator dependent (urine microscopy is further discussed in [Chapter 4](#)). An alternative is to assess urinary RBC size by Coulter counter analysis because dysmorphic RBCs are smaller than normal RBCs, but this method is not useful when RBC numbers in the urine are small. Even in the absence of features of glomerular bleeding, many patients with isolated microhematuria have glomerular disease, most commonly IgA nephropathy or thin basement membrane nephropathy.³⁹ Because there is a low risk for progressive loss of GFR, kidney biopsy in this setting is not usually recommended. Nevertheless, one study showed that microhematuria unexplained by urologic evaluation carries a twofold risk for eventual development of kidney failure,⁴⁰ so these patients should be observed for the development of hypertension, reduced GFR, or proteinuria.

Cyclophosphamide

Past treatment with cyclophosphamide increases the risk of bladder cancer up to ninefold, probably in a dose- and duration-dependent manner. Tumors have been reported 6 to 13 years after cyclophosphamide exposure and are often of high grade. Hematuria is also common after cyclophosphamide in the absence of cancer. If full evaluation does not identify a cause of hematuria, there is no agreed surveillance protocol; it is not clear whether routine follow-up by cystoscopy and urine cytology is valuable, although a high index of suspicion should be maintained.

INVESTIGATION AND MANAGEMENT OF A KIDNEY MASS

The incidence of renal cell carcinoma (RCC) has more than doubled in the last 30 years, now accounting for 3% of all cancers. It is the third most common tumor of the urinary tract but the most lethal. The apparent increased incidence is partly attributed to the widespread use of cross-sectional and ultrasound imaging; more than 50% of new cases are incidental findings on CT, magnetic resonance imaging (MRI), or ultrasound scan.

The primary goal in investigating a kidney mass is to diagnose or exclude malignancy. Ultrasound is 79% sensitive for the detection of kidney parenchymal masses but does not detect lesions that are less than 5 mm. Contrast enhancement with ultrasound also can identify malignancy and improve sensitivity. However, CT scanning before and after contrast administration is the most common modality used to characterize kidney masses. MRI, especially with T2-weighted and diffusion-weighted imaging, may be superior to CT in the correct characterization of benign lesions,⁴¹ but distinguishing an RCC from a benign lesion (i.e., oncocytoma, fat-free angiomyolipoma) is occasionally not feasible radiologically. The choice of imaging techniques and their interpretation are discussed further in [Chapters 5 and 6](#).

Because of the limitations of radiologic evaluation, percutaneous biopsy of a kidney mass is increasingly used to obtain a histologic diagnosis before treatment. Up to 30% of small kidney masses (<3 cm) are nonmalignant. Traditional fears regarding tumor seeding after percutaneous biopsy are not supported by recent studies. Biopsy of a kidney mass is recommended for diagnostic uncertainty despite imaging, before systemic therapy for metastatic RCC and in kidney tumors managed with surveillance or ablative modalities.⁴² Many clinicians also perform a biopsy before embarking on surgery in high-risk or elderly surgical candidates, those who will require technically challenging surgery, and patients with small kidney masses. In a recent systematic review evaluating 57 studies investigating more than 5000 patients, percutaneous biopsy was shown to have a high diagnostic yield, sensitivity (>98%), and specificity (>98%) for kidney cancers, with a low associated complication rate.⁴³

The management of mixed cystic and solid masses is more problematic. [Table 63.6](#) shows the Bosniak classification of cystic kidney masses.⁴⁴ This classification, based on CT appearances, provides the basis for management according to risk for malignancy. The evaluation of multiple cystic lesions in the kidney is discussed further in [Chapters 46 and 47](#).

Tumor size is important; in a large retrospective study of 2935 patients with surgically treated solid kidney tumors, 46% of lesions that were less than 1 cm were benign compared with 22% and 10% of tumors 1 to 2 cm and 4 to 5 cm, respectively. Furthermore, only 2.3% of cancers smaller than 1 cm were of high grade, whereas for tumors greater than 7 cm, the percentage was 58%.⁴⁵ Surveillance studies of small kidney tumors have shown a median growth rate of 0.28 cm per year; about 30% of these lesions will not increase in size.

Traditionally, radical nephrectomy was the gold standard treatment for localized kidney cancers. However, nephron-sparing surgery (NSS) is now recommended in all tumors that are less than 7 cm where technically feasible.⁴² Comparison of these two approaches for T1a (<4 cm) and T1b (4–7 cm) lesions has shown equivalent tumor clearance and a reduction in long-term kidney complications with partial nephrectomy.⁴⁶ Radical nephrectomy therefore should be reserved for larger kidney cancers (≥T2), those technically not amenable to partial nephrectomy, and for patients unlikely to benefit from NSS.⁴²

The last two decades have also seen a shift from open to minimally invasive surgery. Laparoscopic radical nephrectomy and robotic-assisted partial nephrectomy (particularly for small exophytic tumors) have now superseded open approaches for these techniques in most large centers. Minimally invasive treatment modalities give equivalent long-term cancer control, while reducing perioperative blood loss, postoperative analgesic requirement, and hospitalization.

Cryosurgery and radiofrequency ablation via laparoscopic or percutaneous approaches have emerged as treatment options for small localized kidney cancers. There is no consensus on the maximum size of lesions amenable to these techniques, but they are usually

TABLE 63.6 Bosniak Classification and Management of Cystic Kidney Masses^a

Bosniak Class	Features on Imaging	Comment	Management
Class I: simple benign cyst	Round or oval Uniform density <20 HU Unilocular No perceptible wall No contrast enhancement	Majority of asymptomatic cystic lesions	No further intervention required
Class II: benign cyst	One or two nonenhancing septa Calcifications in the wall or septum Hyperdense lesions (50–90 HU, resulting from the presence of blood, protein, or colloid) <3 cm No contrast enhancement		No further intervention required
Class II F: probable benign cyst	Multiple hairline septa “Perceived” enhancement Nodular calcification Hyperdense lesions >3 cm	“Perceived” enhancement resulting from contrast within capillaries of septa	Surveillance with CT scans every 6–12 months
Class III: indeterminate cystic lesions	One or more of the following: Thick, irregular borders Irregular calcifications Thickened or enhancing septa Multilocular form Uniform wall thickening Small nonenhancing nodules	About 40% are neoplastic Magnetic resonance imaging may improve characterization	Surgical exploration
Class IV: presumed malignant cystic masses	Appear malignant Heterogeneous cysts Shaggy, thickened walls or enhancing nodules	Appearances result from necrosis and liquefaction of a solid tumor or a tumor growing in the wall	Surgical exploration

All patients with symptomatic kidney masses should be referred for urologic assessment.

^aApproach to kidney mass found incidentally by ultrasound or CT scanning.

CT, Computed tomography; HU, Hounsfield units.

Data from Israel GM, Bosniak MA. An update of the Bosniak renal cyst classification system. *Urology*. 2005;66:84–88.

reserved for tumors that are less than 4 cm. Such treatments offer the advantages of NSS, especially in patients with high surgical risk. Case series evaluating these modalities individually have reported promising results. However, there are no RCT data on either modality and nonrandomized studies have not shown any significant oncologic or survival benefit when comparing these modalities to partial nephrectomy, although several studies have reported reduced efficacy in terms of local recurrence, metastatic progression, and survival.^{47–49} Further disadvantages of both techniques include absence of histologic confirmation of complete tissue destruction and potential for more difficult salvage surgery for local recurrence.

Adjunctive Therapy for Kidney Cancer

The 5-year survival for patients diagnosed with localized RCC is approximately 80% to 90%. However, in the United States, up to 20% of patients with RCC present with metastasis. The 5-year survival falls to 60% and 10% in the presence of regional and distant metastasis, respectively. Metastatic RCC is resistant to commonly used chemotherapeutic agents, and cytokine therapy with either interleukin (IL)-2 or interferon- α (IFN- α) was standard treatment for metastatic RCC in the United States and Europe until 2005. Although more effective than traditional chemotherapy, it produced only modest response rates with significant toxicity.⁵⁰ Identification of the molecular pathways implicated in the etiology of RCC has led to the development of targeted tyrosine kinase inhibitors and monoclonal antibodies directed against the vascular endothelial growth factor (VEGF) pathway (sorafenib, sunitinib, bevacizumab, pazopanib, axitinib, cabozantinib) and mammalian target of rapamycin (mTOR)

(temsirolimus, everolimus). In randomized studies, many of these agents, compared with standard existing therapy, have improved both progression-free and overall survival when used as first- or second-line therapy.^{51–56} Further development of immune checkpoint inhibitors including nivolumab and pembrolizumab (IgG4 monoclonal antibodies blocking PD-1 [programmed cell death protein 1]) and ipilimumab (CTLA4 [cytotoxic T-lymphocyte-associated protein 4] inhibitor) within the last 5 years have developed treatment further. Landmark trials have demonstrated superior progression-free and overall survival with combination immune check point treatment compared with single VEGF-targeted therapy that had been the first-line treatment only a few years ago.^{57,58} The European Association of Urology has made recommendations on the sequence in which these agents should be used (Table 63.7).⁵⁹ Nephrologic aspects of cancer therapy are further discussed in Chapter 67.

Natural History of Reduced Glomerular Filtration Rate After Surgical Treatment of Kidney Cancer

Normal GFR is usually preserved in the long term after donor transplant nephrectomy. Donors are highly selected to minimize comorbidities and generally are younger than patients treated for kidney tumors. However, emerging long-term evidence suggests those undergoing donor nephrectomy do have increased long-term cardiovascular complications and reduced survival.⁶⁰ Similarly, after surgery for kidney cancer, when the residual functioning kidney tissue is reduced, there is significant risk for late sequelae, including proteinuria, glomerulosclerosis, and progressive kidney failure resulting in increased cardiovascular complications and reduced survival. These risks are

TABLE 63.7 Evidence-Based Recommendations for Targeted Therapy in Metastatic Renal Cell Carcinoma

Prognosis Risk Group ⁵⁹	Standard of Care	Alternative if Unable to Receive/Tolerate Standard of Care
Favorable	Pembrolizumab + axitinib	Sunitinib Pazopanib
Intermediate/ poor	Pembrolizumab + axitinib or Ipilimumab + nivolumab	Cabozantinib Sunitinib Pazopanib

Based on 2020 European Association of Urology Guidelines for renal cell carcinoma.

Data modified from Ljungberg B, Bensalah K, Bex A, et al. European Association of Urology guidelines on Renal Cell Carcinoma, 2020.

www.uroweb.org/guideline/renal-cell-carcinoma/.

thought to be greater when radical rather than partial nephrectomy is performed. A recent systematic review of 21 nonrandomized comparative studies (11,204 patients) evaluating partial nephrectomy versus radical nephrectomy demonstrated superiority for NSS.⁶¹ Partial nephrectomy was associated with greater perioperative blood loss and postoperative complications, affirming that it is a more challenging procedure. However, NSS was associated with a lower long-term decline in eGFR and reduced risk for chronic kidney disease. NSS also provided a more favorable cancer-specific survival. The one RCT to compare partial nephrectomy versus radical nephrectomy also has demonstrated that fewer patients undergoing NSS develop reduced GFR (<60 mL/min/1.73m²), but survival and cancer outcomes are similar with either modality.⁴⁶

SELF-ASSESSMENT QUESTIONS

1. A 50-year-old man presents with right-sided ureteral colic and on subsequent investigation is found to have a 4-mm distal ureteral calculus with no hydronephrosis. He has normal eGFR. In the absence of ongoing pain or sepsis, what is the most appropriate management option?
 - A. Acute endoscopic lasertripsy
 - B. Decompression with ureteric stenting
 - C. Trial of medical expulsive therapy
 - D. Discharge from care of urologist
 - E. Percutaneous nephrostomy
2. A 70-year-old man is seen in an outpatient clinic with a 2-month history of intermittent hematuria. He takes warfarin for atrial fibrillation but is otherwise healthy. His eGFR is in the normal range. What urologic investigations does he require?
 - A. Clotting screen to exclude overanticoagulation as the sole cause of his hematuria
 - B. Clotting screen; proceed to urologic workup only if INR is within therapeutic range
 - C. Clotting screen, flexible cystoscopy, and CT urogram
 - D. Clotting screen, serum creatinine (eGFR), flexible cystoscopy, and CT urogram
3. A 60-year-old man with diabetic nephropathy is referred with a 3-cm lower pole kidney lesion suspicious for malignancy on routine ultrasound scan. These findings are confirmed on CT scan. Despite his comorbidity, he is deemed fit for surgery. What is the most appropriate procedure to consider?
 - A. Open radical nephrectomy to achieve maximum oncologic benefit
 - B. Laparoscopic cryosurgery
 - C. Laparoscopic partial nephrectomy
 - D. Laparoscopic radiofrequency ablation

Renal Cell Carcinoma in von Hippel–Lindau Disease

Von Hippel–Lindau disease (VHL) is a rare autosomal dominant condition with a predisposition to the development of RCC. The genetics, clinical manifestations, and general management of VHL are discussed in Chapter 47. The incidence of RCC in patients with VHL is about 45%. On histologic examination, the tumors are of the clear cell type, often multifocal and bilateral, and can be solid or cystic. The mean age at diagnosis is 39 years, and there is a 30% to 35% risk for tumor progression, metastasis, and death.

Longitudinal imaging studies have described the natural history of VHL kidney lesions. In one study, multifocal lesions were frequent⁶² (on average, 8 per patient), of which 74% were classified as simple cysts, 8% as complex cysts with solid components, and 18% as solid masses. The solid components of VHL lesions almost always contain RCC. During a mean 2.4-year follow-up, most cysts remained the same size (71%) or enlarged (20%), and 9% became smaller. On the contrary, 95% of solid masses increased in size. Although it is generally thought that the cysts are premalignant, the transformation of a simple cyst to a solid lesion was observed in only two patients. Patients with VHL require multidisciplinary management. Surgical intervention is not used for tumors that are less than 3 cm because metastasis is rare below this threshold. In addition, bilateral nephrectomy should be avoided, if possible, because of the substantial morbidity associated with kidney replacement therapy. The standard of care for these patients is partial nephrectomy, and a 10-year survival rate of 81% has been reported.

The results of NSS for VHL appear less satisfactory than for sporadic RCC because of a high risk for local tumor recurrence. Repeated surgery may be needed for new or growing lesions, and for this reason the use of minimally invasive methods is being investigated. Repeated ablation of tumors with radiofrequency and cryotherapy is possible with minimal morbidity; however, the long-term effectiveness of these methods has not yet been established.

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